

Dynamic Modeling Of Northern New York Renewable Energy Sources

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Abstract—A dynamic model of the Northern New York power grid was built using MATLAB Simulink to investigate the potential for control interactions between these generation sources in the North Country and the downstate power grid. The North Country grid primarily consists of wind farms totaling 678.45 MW and the 912 MW St. Lawrence-FDR hydro power plant in Massena, New York. The infrastructure connecting these sources includes the new 345kV transmission lines as well as a 765kV transmission line linked with the power grid in Quebec. Tests were conducted to evaluate grid reliability and potential for power transfer. This included: different types of faults, varying fault duration and simulated outages.

Index Terms—wind, hydro, Northern New York, renewable energy, reliability, dynamic

I. INTRODUCTION

As of 2023, New York State has initiated a plan to achieve a power grid that is 70 percent renewable by 2030 [1]. In order to achieve this goal, they are relying heavily on Northern New York which has significant hydroelectric and wind generation [2]. These renewable sources supply the New York power grid with instabilities, as perfect conditions cannot be secured for their operation [3]. Additionally, due to the relatively small load in Upstate New York there is a large export of power to southern New York [4]. The New York State Power Authority (NYPA) is currently in the process of upgrading the transmission infrastructure to accommodate the addition of new renewable sources and a heavier downstate load [5]. With the added complexity from the renewable sources and new transmission infrastructure, effective modeling is needed to understand potential vulnerabilities in the system. MATLAB Simulink was chosen to develop the model due to its easy to use block format which provides fast tuning of system parameters. The two most critical components of the grid model are six wind farms located near Chateaugay, NY and the St. Lawrence-FDR Power Project hydroelectric dam in Massena, NY as they represent the largest renewable generation in the Northern New York power grid [6]. Another significant component of the North Country grid is the high voltage direct current (HVDC) converter in Chateaugay, Quebec. Northern New York receives a significant import of power from Quebec through this connection [7]. This research presents a MATLAB/Simulink analysis of the Northern New York renewable energy grid. A dynamic model was created in

order to observe instabilities and the response of the grid to any failures.

II. METHODS

In order to quickly develop an overall model of the Northern New York power grid, a set of pre-existing Simulink models for the hydro, wind and HVDC converter were combined together. The HVDC converter and wind farm models were designed by researchers at the Canadian utility Hydro-Québec. The main wind farm model used was the *Synchronous Generator and Full Scale Converter (Type 4) Average Model* shown in figure 1 below [8]. This design includes a number of useful features which enable customization of the model for different wind turbine models. The core component of this model is the synchronous generator which is driven by turbine and drive train control blocks. A DC-DC Boost Converter Average Model steps up the synchronous generator voltage. An average model is utilized which approximates the power electronic components and thus improves simulation performance. An inverter and filter process the boosted DC voltage to AC output on the grid. Within the grid control block, there are voltage and current regulators with gains that can be tuned to enhance system performance.

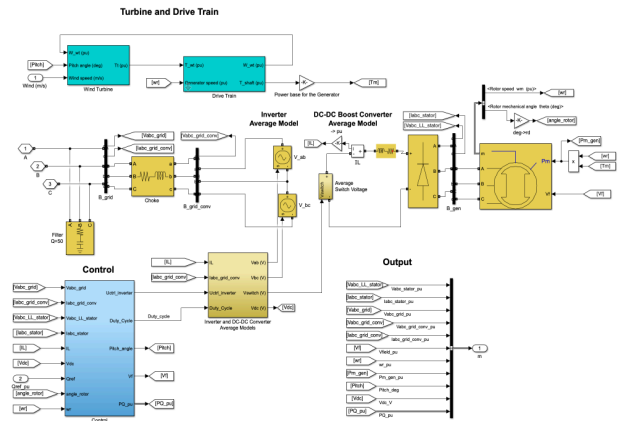


Figure 1 Hydro-Québec Type IV Wind Farm Simulink Model

To decide which wind farms to model, a 2024 map provided by the New York Independent System Operator (NYISO) shown in figure 2 below was used [9]. Six wind farms totalling 678.45 MW near Chateaugay, NY were

selected due to their large capacity and influence over the North Country grid. Next, the United State Geological Survey (USGS) Wind Energy database was used to identify the number of turbines and capacity of each farm [10].

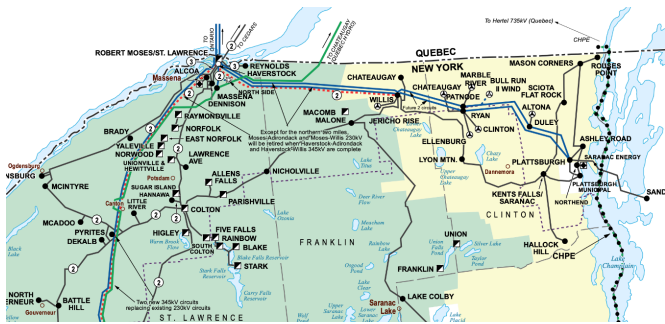


Figure 2 North Country Section of 2024 NYISO New Electric State Electric System Map

In table 1 below, the specifications for each wind farm are listed from the USGS database. The Hydro-Québec model from figure 1 was copied five times for the Type IV wind farms. The nominal power and voltage were changed to match the specifications.

Table 1 Data for Six North Country Wind Farms from USGS Database

Project Name	Turbine Model	Number of Turbines	Turbine Capacity (MW)	Total Capacity (MW)	Turbine Type
Jericho Rise	G114-2.1	37	2.1	77.7	Type III (DFIG)
Marble River	V112-3.075	70	3.075	215.25	Type IV (DFAG)
Noble Altona	GE1.5-77	65	1.5	97.5	Type IV (DFAG)
Noble Chateaugay	GE1.5-77	71	1.5	106.5	Type IV (DFAG)
Noble Clinton	GE1.5-77	67	1.5	100.5	Type IV (DFAG)
Noble Ellenburg	GE1.5-77	54	1.5	81	Type IV (DFAG)

While five of the wind farms are Type IV, Jericho Rise is Type III. For this reason, another model was needed; an induction generator instead of a synchronous one. A similar model shown in figure 3 below from Hydro-Québec was used: *Wind Farm - Doubly Fed Induction Generator* [11]

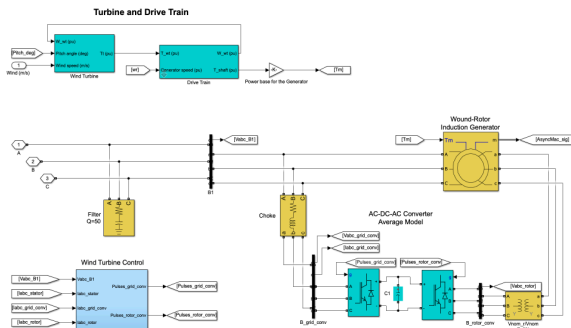


Figure 3 Hydro-Québec Type III Wind Farm Simulink Model

Contained within each wind farm subsystem block are the appropriate transformers to interface with the grid. The first transformer steps the voltage of the turbines (690V) to the collector system voltage of 34.5 kV. The collector line for each wind farm is 1 km which represents an average value of the individual collector lengths. The specific collector line lengths for each turbine could not be modeled as the information was not publicly available. To step the collector voltage to the grid voltage, a 34.5 kV to 345 kV transformer was used. In the case of the Jericho Rise wind farm (type III) an additional transformer was needed, going from 34.5 kV to 115kV before going from 115 kV to 345 kV. For all the transformers, the per unit resistance was increased from 0.002 to 0.0125 and the per unit inductance was decreased from 0.08 to 0.025. While the actual values for the transformers are confidential, these values were chosen to create a more realistic model. For all the wind farm transformers, the base power rating was set to 400 MVA. The largest wind farm (Marble River) is 215.25 MW, so this rating leaves a reasonable safety factor even for the largest wind farm. Figure 4 below shows the distribution infrastructure connected to each wind farm.

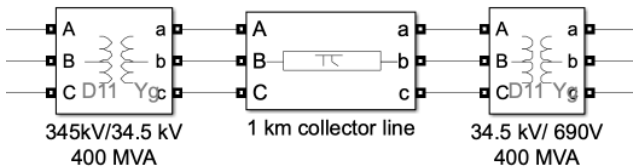
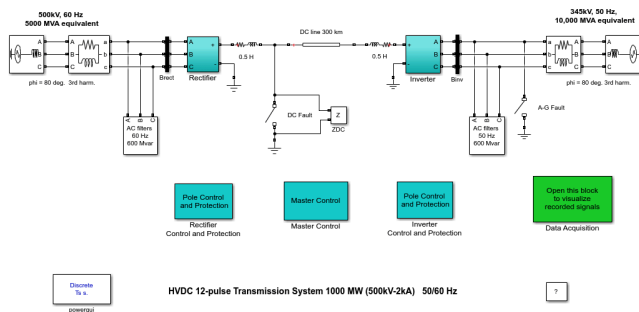


Figure 4 Transformer Configuration for Type IV Wind Farms

The St. Lawrence-FDR Power Project hydro dam was more challenging to model as Hydro-Québec did not have an existing model to use. Instead research from Shewit Tsegaye and Kinde a Fante provided an example hydro turbine model in Simulink [12]. This model makes use of the hydraulic turbine and governor control block and a DC excitation system. The values used in the control and excitation blocks were left at the default values as they perform well in the overall grid model. However, the parameters within the synchronous machine, specifically the time constants and inertia coefficient were increased to provide a more damped response. The hydro dam on the U.S. side has a total of 16

Figure 1: Power Dam Turbines 1-8 block diagram. The diagram shows a three-phase input (A, B, C) connected to a motor model. The motor model has inputs for Stator voltage (v_q), Stator voltage (v_d), Rotor speed (w_m), Rotor speed deviation (die), and Output active power (P_{co}). It also has a gate input. The motor model outputs P_{in} , V_{ref} , P_{ref} , w_{ref} , P_{to0} , d_{ref} , V_{ref} , V_d , V_q , and V_{stab} . The motor model is connected to a three-phase output (A, B, C) and a three-phase input (m, n, p). The motor model is labeled "Power Dam Turbines 1-8".

The model used for the HVDC converter is called *Thyristor Based HVDC Transmission System* [14]. It consists of an inverter, DC lines, a rectifier, and filters. The HVDC converter in Chateauguay, Quebec is back-to-back, meaning there are very short DC connections between the inverter and rectifier. For this reason, the DC line in the Simulink model was removed. Figure 6 below shows the original model with the DC line included.



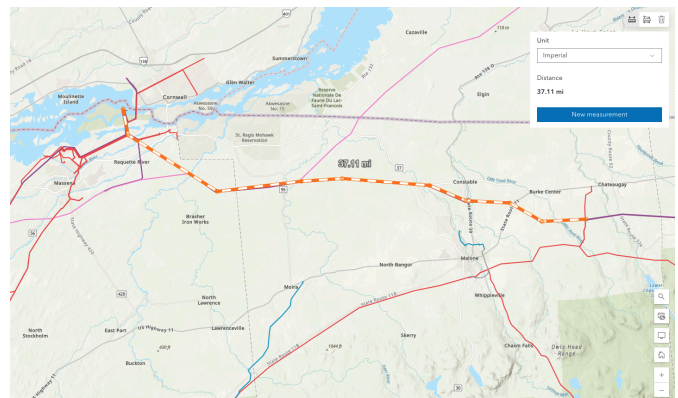
In order to connect each component to form the overall grid, the three phase pi section block was used to model transmission line resistance, inductance and capacitance. Due to the lack of publicly available information, data from a 2019 New York synthetic power grid model was utilized [15]. Developed by University of Wisconsin-Madison, this model contains per unit resistance, reactance and susceptance for each transmission line in New York State. Since NYPA is in the process of upgrading the North Country 230 kV lines to

$$R = \frac{0.00238 \text{ pu} * \frac{(345 \text{ kV})^2}{100 \text{ MVA}}}{62.1 \text{ mi} * 1.609 \text{ km/mi}} = 0.283 \text{ } \Omega/\text{km} \quad (1)$$

$$L = \frac{\frac{0.03627 \text{ pu} * \frac{(345 \text{ kV})^2}{100 \text{ MVA}}}{2\pi * 60}}{62.1 \text{ mi} * 1.609 \text{ km/mi}} = 1.146 \text{ mH/km} \quad (2)$$

$$C = \frac{\frac{1}{(0.93764 \text{ pu} * \frac{(345 \text{ kV})^2}{100 \text{ MVA}}) * 2\pi * 60}}{62.1 \text{ mi} * 1.609 \text{ km/mi}} = 20.9 \text{ nF/km} \quad (3)$$

These calculated values for the 345kV and 765kV lines were used for all sections. Each section length was determined using an ArcGIS database of U.S. Electric Power Lines [16]. The database map includes a measurement feature (shown in figure 7) which allows for individual measurement of each section.



After determining the individual section lengths, they were added to the model. Using another map tool called the Open Infrastructure Database, the main substations and line configurations were identified [17]. For most of the 345 kV

length, there are two sets of lines. Based on the map data, Marble River and Noble Altona are connected to one set and Jericho Rise, Noble Chateaugay, Noble Clinton and Noble Ellenburg are connected to the other set. At each substation, breaker blocks were added, enabling the disconnection of each section of line. Additionally, a bus tie breaker was added to connect the two line sets together. These breakers were later utilized during fault testing. Figure 8 below shows the breaker setup at each substation and figure 9 shows the overall connection of each component in the model.

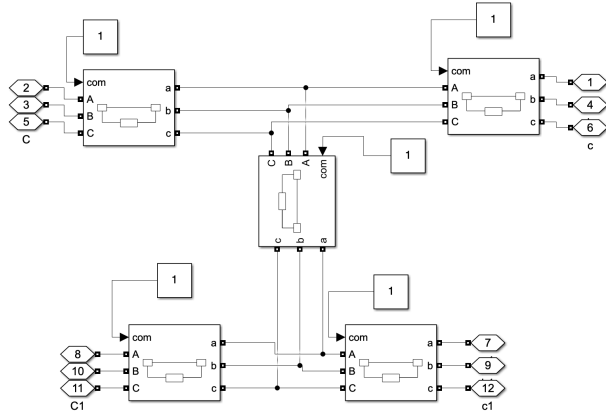


Figure 8 Substation Breaker Configuration

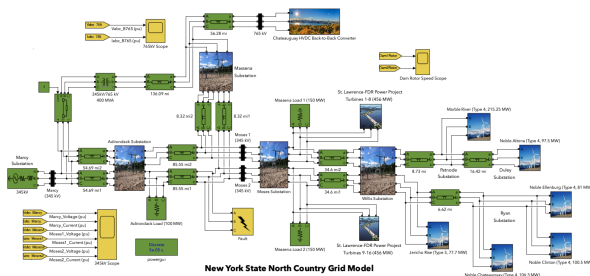


Figure 9 Overall Simulink Model

To simulate loads on the grid, a set of three-phase parallel resistive loads were added at appropriate locations in the model. Two of these loads were added near the Moses substation which represents the load of Massena, NY and the surrounding area. A total value of 300 MW was selected based on the reported energy consumption for the area. It was then divided by two to be distributed on each set of 345 kV lines. For the load farther to the south, a resistive load of 100 MW was added at the Adirondack substation. Like with Massena, NY this value is based on an approximation of the area's energy consumption. The final component to add to the model was the infinite voltage source and load at Marcy substation. Since the model only includes the North Country grid, this substation was selected as the location to stop modeling. The infinite bus provides a convenient simplification of the downstate grid. Default values of 0.8929 Ω and 16.58 mH were used for the impedance.

Fault testing of the dynamic model was conducted to determine the response of the renewable grid to undesirable conditions. The Simulink three-phase fault block was added to the model. In order to test the worst case fault scenario, a full three-phase fault to ground was tested. The total duration tested was primarily at 18 cycles (0.3 seconds) as most faults can be cleared in a real system after this amount of time. However, the system was also tested for longer duration at 60 cycles (1 second). At an 18 cycle duration, each set of assets were disconnected to simulate the effect of different sources going offline. The following combinations were tested: no wind, no hydro, no HVDC, and no 765 kV. The last type of fault testing conducted was each single phase to ground. This was done to simulate a less severe fault.

III. RESULTS

To simulate a base case where all system components were operating, a 3-phase 18 cycle fault at 2 seconds was applied with all breakers connected. In figure 10 below, the sudden drop in voltage is observed with a rapid increase in the current. The maximum overvoltage was about 1.2 pu and the current reached a maximum of 14 pu. However, the system quickly stabilizes after roughly 1 second.

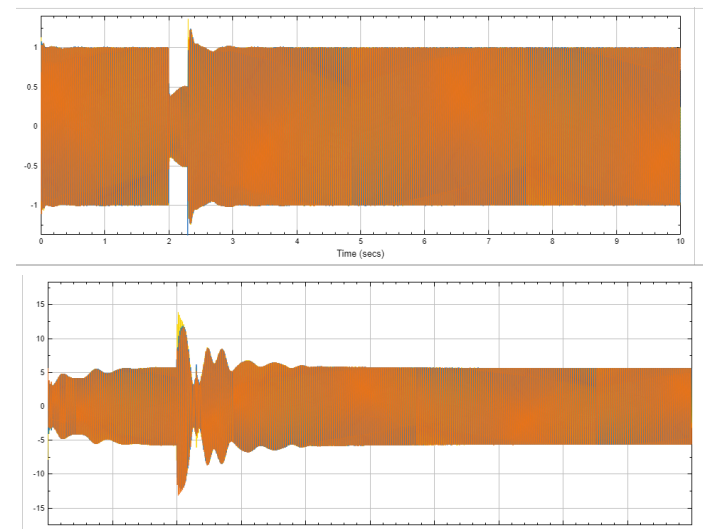


Figure 10 Voltage (pu, top) and Current (pu, bottom) measured at Moses 1 (non-faulted 345 kV line) during 18 cycle fault with all system components online

With adequate system performance at 18 cycles, the next test was conducted with a longer duration of 60 cycles. There was a higher overvoltage of around 1.5 pu and the current did not stabilize from oscillations until after 2 seconds. Additionally, there was significant current decay of around 1 pu.

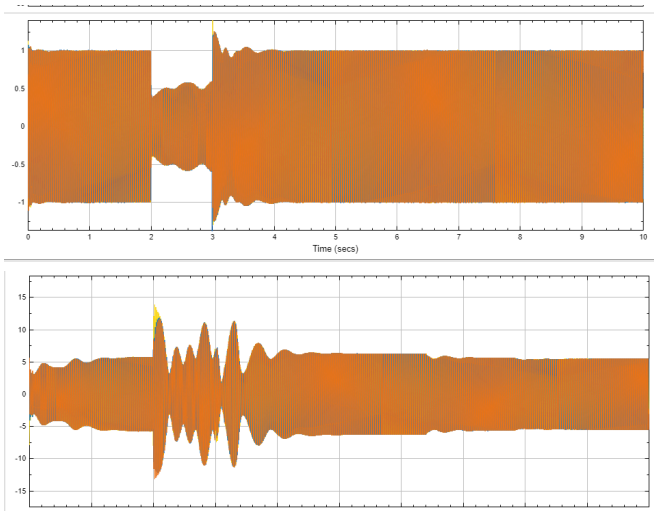


Figure 11 Voltage (pu, top) and Current (pu, bottom) measured at Moses 1 (non-faulted 345 kV line) during 60 cycle fault with all system components online

The fault duration tests revealed the overall resiliency of the system, however there were still considerable oscillations occurring with the current. The primary 2 Hz oscillations are likely caused by the large synchronous generators in the hydro dam, however some subsynchronous oscillations also occur. For this reason, the system was tested without the wind farms. Figure 12 below shows that without them, some of the post fault oscillations are eliminated. It also shows that despite a few oscillations, the wind farms generally do not greatly disrupt the grid operation. The plot looks very similar to the base case with less current due to the disconnected farms.

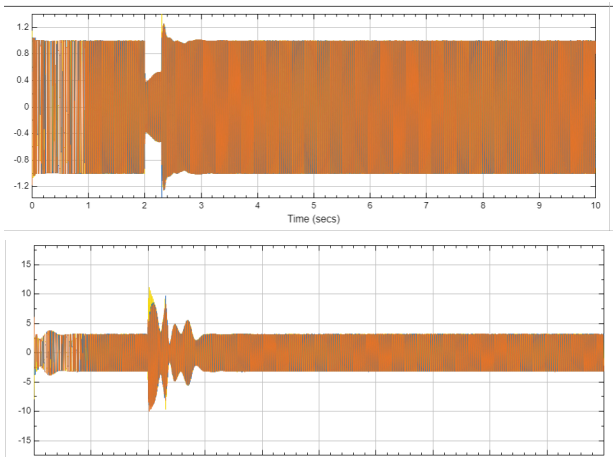


Figure 12 Voltage (pu, top) and Current (pu, bottom) measured at Moses 1 (non-faulted 345 kV line) during an 18 cycle fault with wind farms disconnected from the system.

In addition to the wind farms introducing subsynchronous oscillations, it was also suspected that the HVDC converters were also contributing to them. Another test was run with the HVDC converter to verify this. Figure 13 reveals that the HVDC converters have very little effect on the current after the fault.

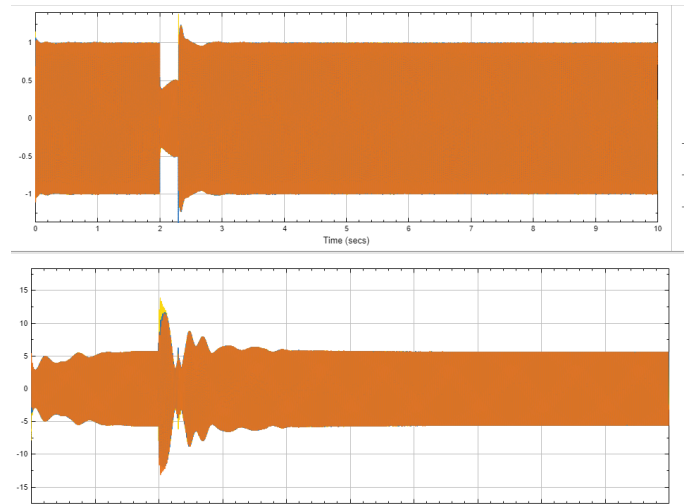


Figure 13 Voltage (pu, top) and Current (pu, bottom) measured at Moses 1 (non-faulted 345 kV line) during an 18 cycle fault with HVDC disconnected from the system.

It was also important to test the absence of hydro generators in the system. One set of the generators on the Moses 1 345kV line were disconnected. This produced significantly less current and the fault current lasted much longer (shown in figure 14). With the lack of a synchronous generator, the system in general takes longer to stabilize. However, there is less overvoltage when compared to the regular test. Overall, it was observed that an outage of hydro dam generators has a greater impact than an outage of wind.

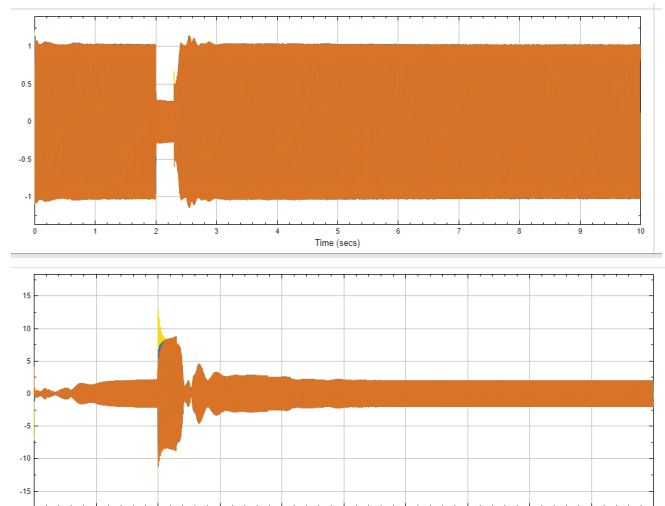


Figure 14 Voltage (pu, top) and Current (pu, bottom) measured at Moses 1 (non-faulted 345 kV line) during an 18 cycle fault with hydro dam turbines 1-8 disconnected from the system.

The last outage test conducted was with the 765 kV transmission lines. They were disconnected at the Massena and Marcy substations. This had a significant effect on the system, making the synchronous generators at the hydro dam unstable. They were unable to stabilize during the entire duration of the test (shown in figure 15). Additional tests were performed with shorter duration faults of 6 and 3 cycles

(0.1 and 0.05 seconds). This did not improve the oscillations, meaning the 765 kV network has a significant effect in stabilizing the 345 kV grid.

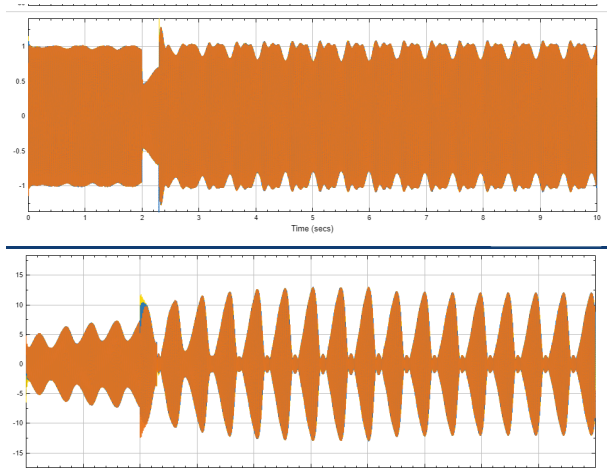


Figure 15 Voltage (pu, top) and Current (pu, bottom) measured at Moses 1 (non-faulted 345 kV line) during an 18 cycle fault with the 765 kV transmission lines disconnected from the system.

In order to eliminate the oscillations produced by the absence of the 765kV transmission lines, the wind farms were also removed from the simulation. Figure 16 demonstrates the only remaining power source left, the hydro power plant, stabilizes the oscillations that were produced by the wind farms. There are no more oscillations in the current before the fault and after the fault the oscillations were damped within a few cycles. From this simulation it was determined that the 765kV transmission lines were responsible for the stabilization of the wind farms in the 345kV lines, and should it ever go offline then the wind farms should also be taken out of the power grid.

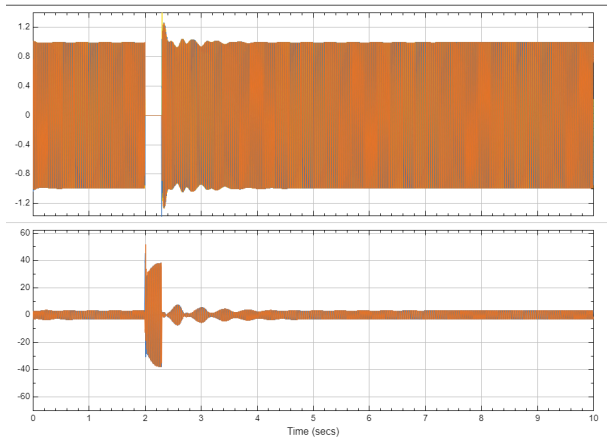


Figure 16 Voltage (pu, top) and Current (pu, bottom) measured at Moses 2 (faulted 345 kV line) during an 18 cycle fault with the 765 kV transmission lines and wind farms disconnected from the system.

Finally, a few tests were run with single phase to ground faults. Figure 17 below shows the case for phase A to ground. Compared to the three phase fault, the overvoltage was similar at around 1.2 pu, but was slower to stabilize. There

were also more oscillations in the current, however they were at a lower magnitude of around 10.5 pu. The single phase tests show that while less fault current will flow, the unbalanced nature of the fault will cause more instability.

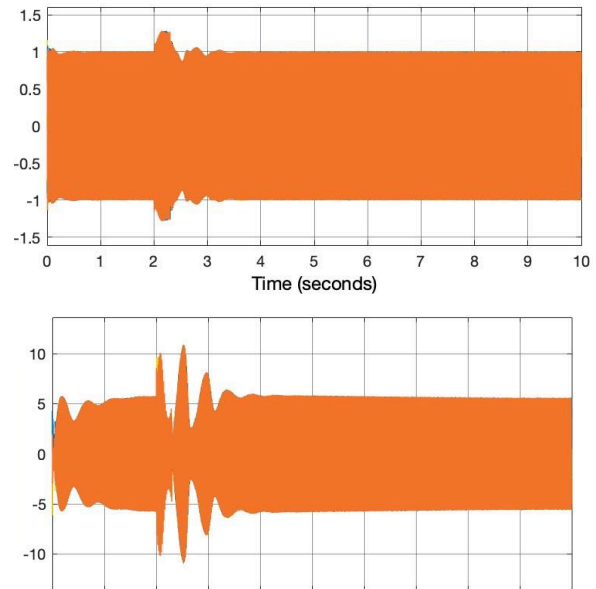


Figure 17 Voltage (pu, top) and Current (pu, bottom) measured at Moses 1 (non-faulted 345 kV line) during an 18 cycle single phase to ground fault with all system components connected.

IV. CONCLUSION

This study involved dynamic modeling of Northern New York in MATLAB Simulink with an emphasis on renewable energy sources. The model was fault tested with varying durations, simulated outages, and different types of faults. The system showed overall resiliency especially to long duration balanced faults. The converter electronics in the HVDC and wind farms had a small effect on the system stability, which means these components have very capable control systems. The hydro dam outage on the other hand had a greater effect when it came to the fault current, so more control of the generator speed would be needed during a fault event. It should be noted that the hydro model used had less control architecture than the wind farms, which may explain this discrepancy.

The greatest impact to the system overall was the outage of the 765 kV transmission lines. While a real world fault is unlikely to completely disrupt the 765 kV transmission network, special attention should be given to preventing damage as the 765 kV network is essential to strengthening the 345 kV network. Finally the unbalanced single phase faults were conducted with the entire system and while there was less fault current that could cause damage, there were longer oscillations which means certain control parameters may need to be altered to stabilize from an unbalanced fault.

Future work would involve improving the accuracy of the hydro dam model. In particular, tuning the control parameters to yield greater stability with a 765 kV

transmission outage. Additionally more experimentation with fault location and breaker control would reveal if there are any weaknesses in the system.

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